Lecture 16

Noncompact Haar Measure and the Modular Function

Objectives. *Lecture 5 integrated over compact groups. This lecture drops compactness. It asks which translation-invariant integrals a Lie group carries and how left and right integration differ. A noncompact Lie group has infinite Haar volume, so the normalised averaging of Lecture 5 is no longer available in general. Every Lie group has a left-invariant and a right-invariant density. With equal normalisations at the identity, the left density divided by the right density is the modular function. It controls right translation, inversion and the right regular representation.*

By the end of this lecture you will be able to:

- Construct the left and right Haar densities of a Lie group, prove that each is unique up to a positive factor, and prove that the total mass is infinite when the group is noncompact.
- Prove the right-translation and inversion identities for $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$.
- Compute both Haar densities and Δ for the affine group $ax + b$.
- Construct the left and right regular representations as strongly continuous unitary representations on $L^2(G)$, and state the integration rules of Part II with their hypotheses.

Lecture 5 built a left-invariant density on a compact Lie group by translating a density at e. That construction did not use compactness. Compactness entered later. It made the integral finite on every continuous function, and it allowed normalisation to total mass 1. It made the image of $j(h) = |\det \mathrm{Ad}(h)|$ a compact subgroup of $(0, \infty)$, so $j \equiv 1$. It gave strict positivity through a finite cover. Without compactness we integrate only compactly supported functions. Strict positivity comes from charts, and j survives as a homomorphism. The left Haar density μ_L below is the density δ of Lecture 5, and the modular function below is $\Delta = j^{-1}$.

Throughout, G is a Lie group of dimension n with Lie algebra $\mathfrak{g} = T_eG$. We write left translation, right translation, conjugation and inversion as

$$L_g(x) = gx, \qquad R_g(x) = xg, \qquad C_g = L_g \circ R_{g^{-1}}, \qquad \iota(x) = x^{-1}.$$

Then $R_g \circ R_h = R_{hg}$ and $d(C_g)_e = \mathrm{Ad}(g)$.

16.1 Densities and integration

Under a linear map T of $\mathbb{R}^n$, Lebesgue measure scales by $|\det T|$. A density imposes this rule on each tangent space.

Definition 16.1. Let V be a real vector space of dimension n. A *positive density* on V is a nonzero function $\delta\colon V^n \to [0,\infty)$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \qquad (T \in \operatorname{End} V).$$

Let $e_1,\dots,e_n$ be a basis and $v_j = Te_j$. Then $\delta(v_1,\dots,v_n) = |\det T|\,\delta(e_1,\dots,e_n)$. So δ is fixed by the number $\delta(e_1,\dots,e_n)$, which is positive since $\delta \neq 0$. Two positive densities on V therefore differ by a positive factor.

Definition 16.2. A *smooth positive density* μ on G assigns a positive density μ_x on T_xG to each $x \in G$ such that, in every chart with coordinates y, the coefficient $m(y) = \mu_x(\partial_1,\dots,\partial_n)$ is smooth.

For a diffeomorphism F of G and a smooth function $\varphi > 0$ put

$$(F^*\mu)_x(v_1,\dots,v_n) = \mu_{F(x)}(dF_x v_1,\dots,dF_x v_n), \qquad (\varphi\mu)_x = \varphi(x)\,\mu_x.$$

The chain rule gives $(F\circ H)^* = H^*F^*$. The definitions give $F^*(\varphi\mu) = (\varphi\circ F)\,F^*\mu$.

Let $C_c(G)$ be the space of continuous functions $f\colon G \to \mathbb{C}$ with compact support.

If $\operatorname{supp} f$ lies in one chart, set $\int_G f\mu = \int_{\mathbb{R}^n} f(y)\,m(y)\,dy$. A change of chart multiplies the coefficient m by the absolute Jacobian determinant. Hence, by the change of variables formula, the integral is well defined.

For general $f \in C_c(G)$ we use a partition of unity, adapted from Lecture 5. Here a *partition of unity* on $\operatorname{supp} f$ is a finite family of smooth functions $\psi_i \geq 0$ on G, each with compact support in one chart, such that $\sum_i \psi_i = 1$ on $\operatorname{supp} f$. On a compact manifold, Lecture 5 requires the sum to equal 1 everywhere. Since f vanishes outside $\operatorname{supp} f$, we have $f = \sum_i \psi_i f$. We add the chart integrals of the functions $\psi_i f$.

We cite this construction and the following three facts. Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII, carries out the same construction with differential forms.

(I1) $f \mapsto \int_G f\mu$ is linear, and $\int_G f\mu > 0$ if $f \geq 0$ and $f \neq 0$.
(I2) $\int_G (f\circ F)\,F^*\mu = \int_G f\mu$ for every diffeomorphism F of G.
(I3) $\int_G f\,(\varphi\mu) = \int_G (f\varphi)\,\mu$.

In a chart, $F^*\mu$ has coefficient $m(F(y))\,|\det DF(y)|$. So (I2) is the change of variables formula.

16.2 Left and right Haar densities

Theorem 16.3 (Haar densities). *Fix a positive density μ_e on $\mathfrak{g}$.*

(a) There is exactly one smooth positive density μ_L with $(\mu_L)_e = \mu_e$ and $L_g^*\mu_L = \mu_L$ for all $g \in G$. Every left-invariant smooth positive density is a positive multiple of μ_L.

(b) There is exactly one smooth positive density μ_R with $(\mu_R)_e = \mu_e$ and $R_g^*\mu_R = \mu_R$ for all $g \in G$. Every right-invariant smooth positive density is a positive multiple of μ_R.

(c) Write $\int_G f(x)\,d_Lx$ for $\int_G f\mu_L$, and similarly d_Rx. For $f \in C_c(G)$ and $g \in G$,

$$\int_G f(gx)\,d_Lx = \int_G f(x)\,d_Lx, \qquad \int_G f(xg)\,d_Rx = \int_G f(x)\,d_Rx.$$

Proof. Uniqueness. Let ν be a left-invariant smooth positive density. Evaluate $L_{x^{-1}}^*\nu = \nu$ at x:

$$\nu_x(v_1,\dots,v_n) = \nu_e\big(d(L_{x^{-1}})_x v_1,\dots,d(L_{x^{-1}})_x v_n\big). \tag{16.1}$$

So ν is determined by ν_e. Two positive densities on $\mathfrak{g}$ differ by a positive factor. Thus $\nu_e = c\mu_e$ for some $c > 0$. Once μ_L is constructed below, the formula gives $\nu = c\mu_L$.

Existence. Define μ_L by (16.1) with $\nu_e = \mu_e$. Linear maps act entrywise on tuples $v = (v_1,\dots,v_n)$. Since $L_{(gx)^{-1}} \circ L_g = L_{x^{-1}}$, the chain rule gives

$$(L_g^*\mu_L)_x(v) = \mu_e\big(d(L_{(gx)^{-1}})_{gx}\,d(L_g)_x v\big) = \mu_e\big(d(L_{x^{-1}})_x v\big) = (\mu_L)_x(v).$$

Smoothness. This step uses the smooth group law. Choose a basis $X_1,\dots,X_n$ of $\mathfrak{g}$. The left-invariant fields $\tilde{X}_i(x) = d(L_x)_e X_i$ are smooth (Lecture 1). Since $L_{x^{-1}} \circ L_x = \mathrm{id}$, we have $d(L_{x^{-1}})_x \tilde{X}_i(x) = X_i$. Hence

$$(\mu_L)_x\big(\tilde{X}_1(x),\dots,\tilde{X}_n(x)\big) = \mu_e(X_1,\dots,X_n) =: c_0 > 0.$$

In a chart write $\tilde{X}_i = \sum_j P_{ji}(y)\,\partial_j$. The matrix $P(y)$ is smooth. It is invertible because $d(L_x)_e$ is an isomorphism. The density rule gives $c_0 = |\det P(y)|\,m(y)$. So $m = c_0\,|\det P|^{-1}$ is smooth.

Right translations. Use $\nu_x(v) = \nu_e(d(R_{x^{-1}})_x v)$. The key identity is $R_{(xg)^{-1}} \circ R_g = R_{x^{-1}}$, since $z\,g\,g^{-1}x^{-1} = zx^{-1}$. The fields $d(R_x)_e X_i$ replace $\tilde{X}_i$.

Integrals. Apply the change of variables (I2) with $F = L_g$ or $F = R_g$. ∎

On a compact group one normalises the total mass to 1, as in Lecture 5. For noncompact G this is impossible, as we now prove. Since the constant function 1 is not in $C_c(G)$, we state the result for compactly supported functions.

Proposition 16.4 (Infinite Haar volume). *Let G be a noncompact Lie group, and let $f \in C_c(G)$ with $f \geq 0$ and $f \neq 0$. There are elements $g_1, g_2, \dots$ of G such that the translates $f_k(x) = f(g_k^{-1}x)$ have pairwise disjoint supports. Hence $F_N = f_1 + \cdots + f_N$ lies in $C_c(G)$, it satisfies $0 \leq F_N \leq \sup f$, and*

$$\int_G F_N\,d_Lx = N\int_G f\,d_Lx, \qquad \int_G f\,d_Lx > 0.$$

Proof. We choose the elements one at a time. Let $S = \operatorname{supp} f$, a compact set, and take $g_1 = e$. Suppose $g_1, \dots, g_{m-1}$ are chosen. The set $\bigcup_{k<m} g_k S S^{-1}$ is compact, as a finite union of continuous images of $S \times S$. Since G is not compact, there is an element g_m outside it.

The support of f_k is $g_k S$. If $g_m S$ met $g_k S$ with $k < m$, then $g_m s = g_k s'$ for some $s, s' \in S$. That gives $g_m = g_k s' s^{-1} \in g_k S S^{-1}$, which contradicts the choice of g_m. So the supports are pairwise disjoint.

At each point at most one term of F_N is nonzero, so $0 \le F_N \le \sup f$. Left invariance in Theorem 16.3(c) gives $\int_G f_k \, d_L x = \int_G f \, d_L x$, and this value is positive by the positivity statement (I1). ∎

So the left Haar integral is unbounded on functions in $C_c(G)$ with values in $[0, \sup f]$, and the total mass of G is infinite. The right Haar integral behaves in the same way, with right translates. A noncompact Lie group therefore has no normalised finite Haar integral, and the averaging arguments of Lecture 5 are no longer available in general.

We use the same density μ_e for both μ_L and μ_R.

16.3 The modular function

Definition 16.5. The *modular function* of G is $\Delta(g) = |\det \operatorname{Ad}(g)|^{-1}$. The group G is *unimodular* if $\Delta \equiv 1$.

Since $\operatorname{Ad} \colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism (Lecture 2), Δ is a smooth homomorphism into the multiplicative group $(0, \infty)$.

Theorem 16.6 (Integration identities)**.** *Let $f \in C_c(G)$ and $g \in G$.*

(i) $R_g^* \mu_L = \Delta(g)\, \mu_L$. *Hence* $\int_G f(xg)\, d_L x = \Delta(g)^{-1} \int_G f(x)\, d_L x$.

(ii) $\mu_R = \Delta^{-1} \mu_L$, *that is,* $\int_G f(x)\, d_R x = \int_G f(x)\, \Delta(x)^{-1}\, d_L x$.

(iii) $\int_G f(x^{-1})\, d_L x = \int_G f(x)\, d_R x$ *and* $\int_G f(x^{-1})\, \Delta(x^{-1})\, d_L x = \int_G f(x)\, d_L x$.

Proof. (i) Left and right translations commute. Hence

$$L_h^* R_g^* \mu_L = (R_g \circ L_h)^* \mu_L = (L_h \circ R_g)^* \mu_L = R_g^* L_h^* \mu_L = R_g^* \mu_L.$$

So $R_g^* \mu_L$ is left-invariant. By the uniqueness in Theorem 16.3(a), it equals $c(g) \mu_L$ for some $c(g) > 0$. We find $c(g)$ at e. Since $L_{g^{-1}} \circ R_g = C_{g^{-1}}$,

$$(R_g^* \mu_L)_e(v) = \mu_e\big(d(L_{g^{-1}})_g \, d(R_g)_e v\big) = \mu_e\big(\operatorname{Ad}(g^{-1}) v\big) = |\det \operatorname{Ad}(g)|^{-1} \mu_e(v).$$

Thus $c(g) = \Delta(g)$. Now the change of variables (I2) with $F = R_g$ and the multiplication rule (I3) give

$$\int_G f \mu_L = \int_G (f \circ R_g)\, R_g^* \mu_L = \Delta(g) \int_G f(xg)\, d_L x.$$

(ii) Use $F^*(\varphi \mu) = (\varphi \circ F) F^* \mu$, the right-translation identity (i) and the homomorphism property of Δ:

$$\big(R_g^* (\Delta^{-1} \mu_L)\big)_x = \Delta(xg)^{-1} \Delta(g)\, (\mu_L)_x = \Delta(x)^{-1} (\mu_L)_x.$$

So $\Delta^{-1}\mu_L$ is right-invariant with value μ_e at e. By uniqueness in Theorem 16.3(b) it equals μ_R. Then the multiplication rule (I3) gives the integral form.

(iii) First, $d\iota_e = -\mathrm{id}$. Indeed, the multiplication map $\mathrm{mult}\colon G\times G\to G$ satisfies $\mathrm{mult}(x,e)=x$ and $\mathrm{mult}(e,x)=x$ for $x\in G$. So $d(\mathrm{mult})_{(e,e)}(v,w)=v+w$ for $v,w\in\mathfrak{g}$. Differentiating $x\,\iota(x)=e$ at e gives $v+d\iota_e v=0$. Therefore

$$(\iota^*\mu_L)_e(v_1,\dots,v_n)=\mu_e(-v_1,\dots,-v_n)=|\det(-\mathrm{id})|\,\mu_e(v_1,\dots,v_n)=\mu_e(v_1,\dots,v_n).$$

Next, $(xg)^{-1}=g^{-1}x^{-1}$ says $\iota\circ R_g=L_{g^{-1}}\circ\iota$. Hence

$$R_g^*\iota^*\mu_L=(\iota\circ R_g)^*\mu_L=(L_{g^{-1}}\circ\iota)^*\mu_L=\iota^*L_{g^{-1}}^*\mu_L=\iota^*\mu_L.$$

By the uniqueness in Theorem 16.3(b), $\iota^*\mu_L=\mu_R$. Apply the change of variables (I2) with $F=\iota$ to $f\circ\iota$. This gives $\int_G f\,d_Rx=\int_G f(x^{-1})\,d_Lx$. For the second identity apply the first to $h=f\Delta\in C_c(G)$ and use the density relation (ii):

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_Lx=\int_G h(x)\,d_Rx=\int_G f(x)\Delta(x)\Delta(x)^{-1}\,d_Lx.$$

∎

Remark 16.7. The right-translation identity (i) is equivalent to $\int_G f(xg^{-1})\,d_Lx=\Delta(g)\int_G f\,d_Lx$. For Borel sets E, using the Radon measure of § 16.6, it reads $\mu_L(Eg)=\Delta(g)\mu_L(E)$. This is Folland's convention. Some texts call $|\det\mathrm{Ad}(g)|$ the modular function. When comparing sources, check which defining identity is used.

Corollary 16.8. *(a) Every compact Lie group is unimodular.*

(b) If G is connected, then $\Delta(\exp X)=e^{-\operatorname{tr}\operatorname{ad}X}$. So G is unimodular if and only if $\operatorname{tr}\operatorname{ad}X=0$ for all $X\in\mathfrak{g}$. This holds when $\mathfrak{g}$ is abelian, nilpotent or semisimple.

Proof. (a) Since G is compact and Δ is continuous, the image $\Delta(G)$ is a compact subgroup of $(0,\infty)$. If $t\neq 1$ lies in it, then so does the unbounded set $\{t^k:k\in\mathbb{Z}\}$. So $\Delta(G)=\{1\}$.

(b) We have $\mathrm{Ad}(\exp X)=e^{\operatorname{ad}X}$ (Lecture 2) and $\det e^{A}=e^{\operatorname{tr}A}$. This gives the formula. If all traces vanish, then $\Delta=1$ on $\exp\mathfrak{g}$. This set contains a neighbourhood U of e (Lecture 2). The subgroup H generated by U is open, since $hU\subset H$ for $h\in H$. It is closed, since its complement is a union of open cosets. By connectedness $H=G$, so $\Delta\equiv 1$. Conversely, $\Delta\equiv 1$ gives $e^{-t\operatorname{tr}\operatorname{ad}X}=1$ for all t, so $\operatorname{tr}\operatorname{ad}X=0$. For nilpotent $\mathfrak{g}$ each $\operatorname{ad}X$ is nilpotent. For semisimple $\mathfrak{g}$ we have $\mathfrak{g}=[\mathfrak{g},\mathfrak{g}]$ and $\operatorname{tr}\operatorname{ad}[X,Y]=\operatorname{tr}[\operatorname{ad}X,\operatorname{ad}Y]=0$. ∎

16.4 The affine group

The *$ax+b$ group* is

$$G=\left\{g(a,b)=\begin{pmatrix}a & b\\ 0 & 1\end{pmatrix}:\ a>0,\ b\in\mathbb{R}\right\}.$$

It acts on $\mathbb{R}$ by $x\mapsto ax+b$. The pair (a,b) is a global chart onto $(0,\infty)\times\mathbb{R}$. So integrals over G are ordinary double integrals. Matrix multiplication gives

$$g(a,b)\,g(a',b')=g(aa',\ ab'+b),\qquad g(a,b)^{-1}=g(a^{-1},-a^{-1}b).$$

A tangent vector at g is a matrix $\begin{pmatrix}\alpha & \beta\\ 0 & 0\end{pmatrix}$, which is $\alpha\partial_a+\beta\partial_b$ in the chart. The Lie algebra has basis

$$X=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix},\qquad Y=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix},\qquad [X,Y]=Y.$$

Translations are restrictions of linear maps on matrices. Hence $d(L_g)_eZ=gZ$ and $d(R_g)_eZ=Zg$. We fix the density μ_e on $\mathfrak{g}$ by $\mu_e(X,Y)=1$.

Left density. For $g=g(a,b)$,

$$gX=\begin{pmatrix}a & 0\\ 0 & 0\end{pmatrix}=a\,\partial_a,\qquad gY=\begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix}=a\,\partial_b.$$

So the matrix P from the smoothness step in the proof of Theorem 16.3 is

$$P=\begin{pmatrix}a & 0\\ 0 & a\end{pmatrix}.$$

Then $m_L=|\det P|^{-1}=a^{-2}$.

Right density. Similarly,

$$Xg=\begin{pmatrix}a & b\\ 0 & 0\end{pmatrix}=a\,\partial_a+b\,\partial_b,\qquad Yg=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}=\partial_b,\qquad P=\begin{pmatrix}a & 0\\ b & 1\end{pmatrix}.$$

Then $m_R=a^{-1}$. In summary,

$$d_Lx=a^{-2}\,da\,db,\qquad d_Rx=a^{-1}\,da\,db.$$

Modular function. With $g^{-1}=\begin{pmatrix}a^{-1} & -a^{-1}b\\ 0 & 1\end{pmatrix}$ we compute

$$gXg^{-1}=\begin{pmatrix}1 & -b\\ 0 & 0\end{pmatrix}=X-bY,\qquad gYg^{-1}=\begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix}=aY.$$

In the basis (X,Y),

$$\mathrm{Ad}(g)=\begin{pmatrix}1 & 0\\ -b & a\end{pmatrix},\qquad \det\mathrm{Ad}(g)=a,\qquad \Delta(g(a,b))=a^{-1}.$$

This matches the density relation in Theorem 16.6(ii): $\Delta^{-1}m_L=a\cdot a^{-2}=a^{-1}=m_R$. It also matches the trace formula in Corollary 16.8(b). Indeed $\operatorname{ad}X$ fixes Y and kills X, so $\operatorname{tr}\operatorname{ad}X=1$. And $\exp tX=g(e^t,0)$ has $\Delta=e^{-t}$. Hence the $ax+b$ group is not unimodular.

The connectedness hypothesis in Corollary 16.8(b) is needed. Let H be the subgroup of elements $g(a,b)$ with $a\in 2^{\mathbb{Z}}$ and $b\in\mathbb{R}$. It is closed, since $2^{\mathbb{Z}}$ is closed in $(0,\infty)$. Its Lie algebra $\mathbb{R}Y$ is abelian. Yet $\mathrm{Ad}(g)Y=gYg^{-1}=aY$ for $g=g(a,b)\in H$. So the modular function Δ_H of H satisfies $\Delta_H(g(a,b))=a^{-1}$, which is not identically 1.

Direct check of the right-translation identity in Theorem 16.6(i). Let $x=g(a',b')$ and $g=g(a,b)$. Then $xg=g(a'a,\ a'b+b')$. Substitute $u=a'a$, $v=a'b+b'$. Then $du\,dv=a\,da'\,db'$ and $a'^{-2}=a^2u^{-2}$. So

$$\int_G f(xg)\,d_Lx=\iint f(a'a,\ a'b+b')\,\frac{da'\,db'}{a'^2}=\iint f(u,v)\,\frac{a^2}{u^2}\,\frac{du\,dv}{a}=a\int_G f\,d_Lx.$$

This is $\Delta(g)^{-1}\int_G f\,d_Lx$.

Direct check of the second inversion identity in Theorem 16.6(iii). Here $x^{-1}=g(1/a,-b/a)$ and $\Delta(x^{-1})=a$. Substitute $u=1/a$, $v=-b/a$. The Jacobian determinant of $(a,b)\mapsto(u,v)$ is

$$\det\begin{pmatrix}-a^{-2} & 0\\ ba^{-2} & -a^{-1}\end{pmatrix}=a^{-3},\qquad\text{so}\qquad da\,db=u^{-3}\,du\,dv.$$

Since $a\cdot a^{-2}=a^{-1}=u$,

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_Lx=\iint f\big(\tfrac{1}{a},-\tfrac{b}{a}\big)\,\frac{a\,da\,db}{a^2}=\iint f(u,v)\,u\,u^{-3}\,du\,dv=\int_G f\,d_Lx.$$

Translating a rectangle

In the $ax+b$ group the modular function can be computed from weighted areas, without differentiating a matrix. For a region E in the (a,b)-chart we write

$$\mu_L(E)=\iint_E a^{-2}\,da\,db$$

and call it the *left Haar measure* of E. Now let

$$E=\{(a,b):1\le a\le 2,\ 0\le b\le 1\}.$$

Its left Haar measure is

$$\mu_L(E)=\int_1^2\int_0^1 a^{-2}\,db\,da=\big[-a^{-1}\big]_1^2=\frac{1}{2}.$$

For $h=g(s,t)$, right translation sends (a,b) to $(as,b+at)$. Its Jacobian in (a,b) is s, while the new density coefficient is $(as)^{-2}$. Hence

$$\mu_L(Eh)=\int_E (as)^{-2}s\,da\,db=s^{-1}\mu_L(E)=\Delta(h)\mu_L(E).$$

The shear parameter t changes the shape of the region but not this scale factor. Left translation by h has Jacobian s^2 and density coefficient $(sa)^{-2}$, so it preserves $\mu_L(E)$.

An integrand $f(xh)$ involves a third region. The function $x\mapsto \mathbf{1}_E(xh)$ equals 1 exactly when $xh\in E$, so it is the indicator of Eh^{-1}. Figure 16.1 draws the three regions for $h=g(2,1)$. Here $hE=[2,4]\times[1,3]$, and Eh is the parallelogram with vertices $(2,1)$, $(4,2)$, $(4,3)$ and $(2,2)$. The region Eh^{-1} is the set of (a,b) with $\tfrac12\le a\le 1$ and $-a\le b\le 1-a$.

The weighted masses are

$$\mu_L(hE)=\tfrac12=\mu_L(E),\qquad \mu_L(Eh)=\tfrac14=\Delta(h)\mu_L(E),\qquad \mu_L(Eh^{-1})=1=\Delta(h)^{-1}\mu_L(E).$$

The inverse in the region Eh^{-1} explains why the integral identity in Theorem 16.6(i) has $\Delta(h)^{-1}$, whereas the region identity has $\Delta(h)$.

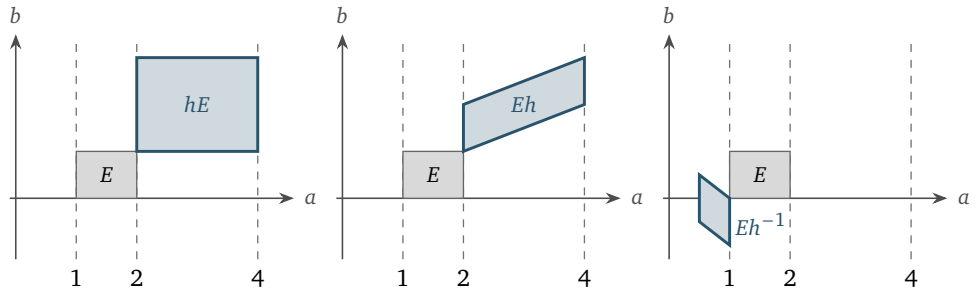


Figure 16.1: The rectangle E in the (a,b) half-plane, in grey, with the left translate hE, the right translate Eh and the region Eh^{-1} for $h = g(2,1)$. The measure of a region is the integral of $a^{-2}\,da\,db$, not its Euclidean area.

16.5 Regular representations

Definition 16.9. The space $L^2(G)$ is the Hilbert completion of $C_c(G)$ for the inner product

$$\langle f, h\rangle = \int_G \overline{f(x)}\, h(x)\, d_L x.$$

By the positivity statement (I1), this inner product is positive definite. For $f \in C_c(G)$ define

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = \Delta(g)^{1/2} f(xg).$$

Lemma 16.10. *Let $f \in C_c(G)$ and $\varepsilon > 0$. There is a neighbourhood $V = V^{-1}$ of e with $|f(yx) - f(x)| < \varepsilon$ and $|f(xy) - f(x)| < \varepsilon$ for all $x \in G$ and $y \in V$.*

Proof. We treat $f(yx)$. The other case is symmetric. If U_L works for $f(yx)$ and U_R for $f(xy)$, put $V' = U_L \cap U_R$ and $V = V' \cap V'^{-1}$.

Fix a compact neighbourhood W of e. Choose a decreasing neighbourhood basis $V_1 \supset V_2 \supset \cdots$ of e inside W. Suppose no V_k works. Then there are $y_k \in V_k$ and x_k with $|f(y_k x_k) - f(x_k)| \geq \varepsilon$. One of x_k and $y_k x_k$ lies in $K = \operatorname{supp} f$. So x_k lies in the compact set $K \cup W^{-1}K$. Since G is second countable and Hausdorff, compact subsets are sequentially compact. So a subsequence has $x_k \to x$. As $y_k \to e$, also $y_k x_k \to x$. Continuity gives $f(y_k x_k) - f(x_k) \to 0$, which is a contradiction. ∎

Proposition 16.11. *The maps λ and ρ extend to strongly continuous unitary representations of G on $L^2(G)$, and $\lambda(g)\rho(h) = \rho(h)\lambda(g)$ for all $g, h \in G$.*

Proof. Both are homomorphisms on $C_c(G)$. For example,

$$\rho(g)\rho(h)f(x) = \Delta(g)^{1/2}\Delta(h)^{1/2} f(xgh) = \rho(gh)f(x).$$

Both operators preserve norms. By the left invariance in Theorem 16.3(c) and the right-translation identity in Theorem 16.6(i), applied to $|f|^2$,

$$\|\lambda(g)f\|^2 = \int_G |f(g^{-1}x)|^2\, d_L x = \|f\|^2, \qquad \|\rho(g)f\|^2 = \Delta(g)\int_G |f(xg)|^2\, d_L x = \|f\|^2.$$

Each operator is a bijection of $C_c(G)$, with inverse $\lambda(g^{-1})$ or $\rho(g^{-1})$. An isometric bijection of a dense subspace extends to an isometry with closed, dense range, hence to a unitary operator. The homomorphism property passes to the extensions by continuity. The operators commute, by the computation

$$\lambda(g)\rho(h)f(x) = \Delta(h)^{1/2} f(g^{-1}xh) = \rho(h)\lambda(g)f(x).$$

Strong continuity. Fix $f \in C_c(G)$ with support K, and a compact neighbourhood $W = W^{-1}$ of e. For $g \in W$, the functions $\lambda(g)f - f$ and $f(\cdot\, g) - f$ vanish outside $WK \cup KW$. A finite sum of chart bump functions gives $\varphi \in C_c(G)$ with $\varphi \geq 0$ and $\varphi \geq 1$ on $WK \cup KW$. Fix $\varepsilon > 0$. Take the neighbourhood supplied by the uniform-continuity statement of Lemma 16.10 and intersect it with W to obtain $V \subset W$. For $g \in V$,

$$\|\lambda(g)f - f\|^2 \leq \varepsilon^2 \int_G \varphi \, d_L x.$$

The function φ was fixed before ε, and $\varepsilon > 0$ is arbitrary. Hence $\lambda(g)f \to f$ as $g \to e$. For ρ use

$$\|\rho(g)f - f\| \leq \Delta(g)^{1/2} \, \|f(\cdot\, g) - f\| + |\Delta(g)^{1/2} - 1| \, \|f\|$$

and the continuity of Δ. The squared norm $\|f(\cdot\, g) - f\|^2$ obeys the same bound as $\|\lambda(g)f - f\|^2$, by the second estimate of Lemma 16.10. Hence $\rho(g)f \to f$ as $g \to e$.

Thus λ and ρ are continuous at e on the dense subspace $C_c(G)$. The dense-set criterion for strong continuity in Lecture 15 extends this to every vector of $L^2(G)$ and every point of G. ∎

Without the factor $\Delta(g)^{1/2}$, right translation multiplies squared norms by $\Delta(g)^{-1}$. On the $ax + b$ group the factor $\Delta(g)^{-1}$ equals a for $g = g(a, b)$. Exercise 16.5 shows that λ and ρ are unitarily equivalent.

We test this on the rectangle E of the $ax + b$ group, with $h = g(s, t)$ as in § 16.4. For $f \in C_c(G)$, the right-translation identity in Theorem 16.6(i), applied to $|f|^2$, gives $\|f(\cdot\, h)\|_2^2 = \Delta(h)^{-1}\|f\|_2^2 = s\,\|f\|_2^2$.

The indicator $\mathbf{1}_E$ of E is not continuous, so it is not in $C_c(G)$. It defines a vector of $L^2(G)$, as follows. For $0 < \delta \leq \frac{1}{2}$ let $\varphi_\delta(a, b) = u_\delta(a)v_\delta(b)$ be a product of trapezoid functions. The function u_δ equals 1 on $[1, 2]$, vanishes outside $[1 - \delta, 2 + \delta]$, and is linear in between. The function v_δ is built in the same way on $[0, 1]$. Then $\varphi_\delta \in C_c(G)$ and $0 \leq \varphi_\delta \leq 1$. It equals $\mathbf{1}_E$ except on a region of Euclidean area $4\delta + 4\delta^2$, and $a^{-2} \leq 4$ on this region. Hence $\int_G |\mathbf{1}_E - \varphi_\delta|^2 \, d_L x \leq 4(4\delta + 4\delta^2) \to 0$. So the functions φ_δ form a Cauchy family in $L^2(G)$, whose limit is the vector $\mathbf{1}_E$, with $\|\mathbf{1}_E\|_2^2 = \mu_L(E)$. Applying the identity above to φ_δ and passing to the limit gives

$$\|\mathbf{1}_E(\,\cdot\, h)\|_2^2 = \mu_L(Eh^{-1}) = s\,\mu_L(E).$$

Multiplication by $\Delta(h)^{1/2} = s^{-1/2}$ restores the norm in both cases.

16.6 Locally compact groups

A topological group is *locally compact* if e has a compact neighbourhood. We cite the following result (Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2). Let G be a locally compact Hausdorff group. A linear functional I on $C_c(G)$ is *positive* if $I(f) \geq 0$ whenever $f \geq 0$. There is a nonzero positive linear functional I on $C_c(G)$ with $I(f(g\,\cdot)) = I(f)$ for all $f \in C_c(G)$ and $g \in G$. It is unique up to a positive factor. There is a unique continuous homomorphism $\Delta\colon G \to (0,\infty)$ satisfying $I(f(\cdot\, g^{-1})) = \Delta(g) I(f)$.

A general locally compact group has no normalisation at e that compares left and right integrals. The cited result instead contains two identities. They replace the density relation in Theorem 16.6(ii) and the inversion identities in part (iii). First, the functional $f \mapsto I(f\Delta^{-1})$ is right-invariant. Second, I takes the same value on $f \in C_c(G)$ and on the function $x \mapsto f(x^{-1})\Delta(x^{-1})$.

The Riesz representation theorem realises I as a Radon measure. We cite Folland, *Real Analysis*, second edition, Theorem 7.2 in §7.1.

For a Lie group, uniqueness makes I a multiple of $f \mapsto \int_G f\, d_L x$. So the two modular functions agree. Since $C_c(G)$ is dense in L^2 of a Radon measure (Folland, *Real Analysis*, Proposition 7.9 in §7.2), our $L^2(G)$ is the usual one, in the sense made precise below.

Integration rules for Part II

We now collect the measure theory used in the rest of Part II. We cite it from Folland, *Real Analysis*, second edition, Chapters 2, 6 and 7, and we do not prove it. Let X be a Lie group, a quotient G/H of a Lie group by a closed subgroup, or an open subset of $\mathbb{R}^n$. Such a space is locally compact and second countable, and it is a countable union of compact sets. We state the rules only for these spaces. General locally compact groups need further hypotheses in the product theorems.

A *Borel set* of X is a member of the σ-algebra generated by the open sets. A function $f\colon X \to \mathbb{C}$ is *measurable* if the preimage of every open set is a Borel set. A *positive Radon measure* μ on X assigns a value in $[0,\infty]$ to each Borel set. It is countably additive and finite on compact sets. On our spaces $\mu(E)$ is the infimum of $\mu(U)$ over open $U \supset E$, and the supremum of $\mu(K)$ over compact $K \subset E$. A *null set* is a Borel set of measure 0, or a subset of one. A property holds *almost everywhere* if it fails only on a null set.

The word “positive” says only that the values are nonnegative. The measure μ has *full support* if $\mu(U) > 0$ for every nonempty open set U. Equivalently, $\int_X \varphi\, d\mu > 0$ for every nonzero $\varphi \in C_c(X)$ with $\varphi \geq 0$. For example, the point mass at e is a positive Radon measure without full support. Haar measures have full support by the positivity statement (I1), and so do the quotient measures of Lecture 17. Lecture 25 also uses positive measures without full support.

Suppose that μ has full support. Then $\int_X \bar{\varphi}\psi\, d\mu$ is an inner product on $C_c(X)$, and its Hilbert completion is the usual space $L^2(X,\mu)$. Its vectors are the measurable f with $\int_X |f|^2\, d\mu < \infty$, and two functions define one vector when they agree almost everywhere. On $\mathbb{R}$, the elements of the completion $L^2(\mathbb{R})$ of Lecture 15 are therefore classes of functions of this kind.

A measurable f is *integrable* if $\int_X |f|\,d\mu < \infty$. Let μ and ν be positive Radon measures on such spaces X and Y. We use the following rules.

- *Tonelli.* If $F\colon X \times Y \to [0,\infty]$ is measurable, then the two iterated integrals of F are equal. Their common value may be infinite.
- *Fubini.* If $F\colon X \times Y \to \mathbb{C}$ is measurable and one iterated integral of $|F|$ is finite, then the two iterated integrals of F exist, are finite and are equal. We use Tonelli to check the hypothesis.
- *Dominated convergence.* Let f_j be measurable with $f_j \to f$ almost everywhere. Suppose there is one integrable function g with $|f_j| \le g$ almost everywhere for all j. Then f is integrable and $\int_X f_j\,d\mu \to \int_X f\,d\mu$.
- *Differentiation under the integral.* Let $I \subset \mathbb{R}$ be an open interval. Let $F(s,\cdot)$ be integrable for each $s \in I$, and let $s \mapsto F(s,x)$ be differentiable for each $x \in X$. Suppose there is one integrable function g with $|\partial_s F(s,x)| \le g(x)$ for all $s \in I$ and $x \in X$. Then $\frac{d}{ds}\int_X F(s,x)\,d\mu(x) = \int_X \partial_s F(s,x)\,d\mu(x)$. Indeed, the mean value inequality bounds the difference quotients by g, and dominated convergence applies to them.
- *Change of variables.* Let $\Phi\colon U \to V$ be a C^1 diffeomorphism of open subsets of $\mathbb{R}^n$, and let $f\colon V \to \mathbb{C}$ be integrable, or measurable with values in $[0,\infty]$. Then $\int_V f(y)\,dy = \int_U f(\Phi(x))\,|\det D\Phi(x)|\,dx$.

The common bound in dominated convergence cannot be dropped. On $\mathbb{R}$ the functions $j\,\mathbf{1}_{(0,1/j)}$ tend to 0 everywhere, but each has integral 1. For the differentiation rule, take $F(s,t) = e^{-t^2+ist}$ on $\mathbb{R}$. Then $|\partial_s F(s,t)| = |t|e^{-t^2}$, which is integrable and does not depend on s. So $\int_{\mathbb{R}} e^{-t^2+ist}\,dt$ may be differentiated under the integral.

For Tonelli and Fubini, here is the bound that Lecture 18 uses. Let $f, h \in L^2(\mathbb{R})$ and $\lambda \neq 0$. Tonelli lets us integrate over t first. Cauchy–Schwarz and translation invariance bound the inner integral by $\|h\|\,\|f\|$. The two Gaussian integrals are $2\sqrt{\pi}$ and $2\sqrt{\pi}/|\lambda|$. Hence

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\,|f(t+a)|\,dt\,da\,db \le \frac{4\pi}{|\lambda|}\,\|h\|\,\|f\|.$$

This finite bound is the hypothesis of Fubini for the same integrand without absolute values.

Self-checks

1. In the $ax+b$ group let $g = g(2,5)$. Suppose $\int_G f\,d_L x = 7$. Find $\Delta(g)$ and $\int_G f(xg)\,d_L x$.

   Answer. $\Delta(g) = 1/2$. By the right-translation identity in Theorem 16.6(i), the integral is $\Delta(g)^{-1} \cdot 7 = 14$.

2. Suppose some left-invariant smooth positive density on G is also right-invariant. Must G be unimodular?

   Answer. Yes. By the uniqueness in Theorem 16.3(a), the density is $c\mu_L$ for some $c > 0$, so $R_g^*\mu_L = \mu_L$ for all $g \in G$. By the right-translation identity in Theorem 16.6(i), this means $\Delta(g) = 1$ for all $g \in G$.

3. On the $ax+b$ group, by what factor does $f \mapsto f(\cdot\, g(3,1))$ multiply $\|f\|^2$ in $L^2(G)$?

   Answer. By the right-translation identity in Theorem 16.6(i), applied to $|f|^2$, the factor is $\Delta(g(3,1))^{-1} = 3$. Hence the map is not unitary without the factor $\Delta^{1/2}$.

Summary

Translating a fixed positive density on $\mathfrak{g}$ gives the left and right Haar densities. Each is unique up to a positive factor. On a noncompact Lie group the total mass is infinite, so no normalised finite Haar integral exists. Right translation scales μ_L by $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. When μ_L and μ_R have the same value at e, we have $\mu_R = \Delta^{-1}\mu_L$, and inversion exchanges μ_L and μ_R. For the $ax + b$ group, $d_L x = a^{-2}\,da\,db$, $d_R x = a^{-1}\,da\,db$ and $\Delta = a^{-1}$. The factor $\Delta^{1/2}$ makes right translation unitary on $L^2(G)$. Haar measures and quotient measures are positive Radon measures with full support, and the cited integration rules of § 16.6 hold in the second-countable spaces of Part II.

The main groups of Part II are unimodular: $\mathbb{R}^n$, the Heisenberg group, $\mathrm{SL}(2,\mathbb{R})$, and the Lorentz and Poincaré groups. Their closed subgroups often are not. For example, the upper triangular subgroup of $\mathrm{SL}(2,\mathbb{R})$ with positive diagonal is an $ax + b$ group (Exercise 20.6). Lecture 17 compares Δ_G with Δ_H to build unitary representations.

Reading

- Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII “Integration”, sections “Differential Forms and Measure Zero” and “Haar Measure for Lie Groups”. Compare its normalisation of the modular function with ours.
- Kirillov, *Elements of the Theory of Representations*, §9, from the course reference route. Check its modular-function convention.
- Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2, sections “Haar Measure” and “The Modular Function”, for general locally compact groups.